A Formal Bound on the Suboptimality of zCDP Accounting under Heterogeneous Composition

1 Problem Setup

Consider a heterogeneous composition of mechanisms

$$M = (M_1, \dots, M_n),$$

where each mechanism uses the same noise family ν, possibly with different scales or sensitivities. Suppose that each M_i satisfies ρ_i-zCDP, so that the usual zCDP composition theorem gives the total privacy parameter

$$\rho = \sum_{i=1}^{n} \rho_i.$$

The goal is to quantify the suboptimality of this scalar accounting rule relative to the exact Renyi privacy curve.

Throughout this note, we use the convention that a mechanism is ρ-zCDP if, for every order $\alpha > 1$,

$$D_\alpha(P\|Q) \le \rho\alpha,$$

where P, Q are the output laws of the mechanism on neighboring datasets. Under this normalization, define the zCDP accounting defect at order α by

$$\Gamma_{n,\nu}(\alpha) := \rho\alpha - D_\alpha(P\|Q).$$

Equivalently, because Renyi divergence tensorizes over product distributions, one may also write

$$\Gamma_{n,\nu}(\alpha) := \alpha \sum_{i=1}^{n} \rho_i - \sum_{i=1}^{n} D_\alpha(P_i\|Q_i).$$

The purpose of the results below is to give lower bounds on $\Gamma_{n,\nu}(\alpha)$ as a function of the number of compositions n, the allocation $(\rho_1, \dots, \rho_n)$, and the noise law ν.

2 Assumptions

We work under the following standing assumptions.

Assumption 1 (Independent additive composition)**.** *For each coordinate i, the two output laws of M_i on a neighboring pair of datasets are*

$$P_i = \mathrm{Law}(Z_i + \theta_i), \qquad Q_i = \mathrm{Law}(Z_i),$$

where $Z_i \sim \nu_{s_i}$, and the random variables $Z_1, \dots, Z_n$ are independent. The distributions ν_{s_i} belong to a common noise family ν, possibly with different scales s_i. Thus the joint output laws are

$$P = \bigotimes_{i=1}^{n} P_i, \qquad Q = \bigotimes_{i=1}^{n} Q_i.$$

Assumption 2 (Absolute continuity and finite Renyi divergences)**.** *For every coordinate i, assume*

$$P_i \ll Q_i, \qquad Q_i \ll P_i,$$

and

$$D_\alpha(P_i \| Q_i) < \infty \qquad \text{for all } \alpha > 1.$$

Remark 1. *The absolute-continuity and finiteness assumptions hold for many smooth additive noise families with everywhere positive densities. Centering and symmetry of ν are not needed for the tensorization identity, but they are useful when one wants local expansions of the per-coordinate Renyi curve in terms of the shift size. In particular, symmetry typically removes odd-order terms in the small-shift expansion.*

3 Exact Tensorization of the Renyi Curve

Lemma 1 (Tensorization of Renyi divergence)**.** *Under Assumptions 1 and 2, for every $\alpha > 1$,*

$$D_\alpha(P \| Q) = \sum_{i=1}^{n} D_\alpha(P_i \| Q_i).$$

Proof. By definition,

$$D_\alpha(P \| Q) = \frac{1}{\alpha - 1} \log \int \left(\frac{\mathrm{d}P}{\mathrm{d}Q} \right)^{\alpha} \mathrm{d}Q.$$

Since $P = \bigotimes_i P_i$ and $Q = \bigotimes_i Q_i$, we have

$$\frac{\mathrm{d}P}{\mathrm{d}Q}(x_1, \dots, x_n) = \prod_{i=1}^{n} \frac{\mathrm{d}P_i}{\mathrm{d}Q_i}(x_i).$$

Therefore,

$$\begin{aligned} \int \left(\frac{\mathrm{d}P}{\mathrm{d}Q} \right)^{\alpha} \mathrm{d}Q &= \int \prod_{i=1}^{n} \left(\frac{\mathrm{d}P_i}{\mathrm{d}Q_i}(x_i) \right)^{\alpha} \prod_{i=1}^{n} \mathrm{d}Q_i(x_i) \\ &= \prod_{i=1}^{n} \int \left(\frac{\mathrm{d}P_i}{\mathrm{d}Q_i} \right)^{\alpha} \mathrm{d}Q_i. \end{aligned}$$

Taking logarithms and dividing by $\alpha - 1$ gives

$$D_\alpha(P \| Q) = \sum_{i=1}^{n} D_\alpha(P_i \| Q_i),$$

as claimed. □

4 The Exact zCDP Accounting Defect

For each coordinate define its exact zCDP radius by

$$\bar{\rho}_i := \sup_{\beta>1} \frac{D_\beta(P_i\|Q_i)}{\beta}.$$

If a mechanism is certified using a possibly non-minimal zCDP parameter ρ_i, then necessarily

$$\rho_i \geq \bar{\rho}_i.$$

Theorem 1 (Exact zCDP accounting defect under heterogeneous composition)**.** *Under Assumptions 1 and 2, let $P = \bigotimes_i P_i$ and $Q = \bigotimes_i Q_i$. Suppose each coordinate is certified as ρ_i-zCDP, where $\rho_i \geq \bar{\rho}_i$. Then for every $\alpha > 1$,*

$$\Gamma_{n,\nu}(\alpha) := \alpha \sum_{i=1}^{n} \rho_i - D_\alpha(P\|Q)$$

satisfies the exact identity

$$\Gamma_{n,\nu}(\alpha) = \alpha \sum_{i=1}^{n} (\rho_i - \bar{\rho}_i) + \sum_{i=1}^{n} \Big\{ \alpha\bar{\rho}_i - D_\alpha(P_i\|Q_i) \Big\}.$$

In particular,

$$\Gamma_{n,\nu}(\alpha) \geq 0.$$

If each coordinate is calibrated to its exact zCDP radius, namely $\rho_i = \bar{\rho}_i$, then

$$\Gamma_{n,\nu}(\alpha) = \sum_{i=1}^{n} \Big\{ \alpha\rho_i - D_\alpha(P_i\|Q_i) \Big\}.$$

Moreover, $\Gamma_{n,\nu}(\alpha) = 0$ for every allocation $(\rho_1, \ldots, \rho_n)$ and every $\alpha > 1$ if every coordinate has a linear Gaussian-type Renyi curve

$$D_\alpha(P_i\|Q_i) = \alpha\rho_i \qquad \text{for all } \alpha > 1.$$

Proof. By Lemma 1,

$$D_\alpha(P\|Q) = \sum_{i=1}^{n} D_\alpha(P_i\|Q_i).$$

Therefore,

$$\begin{aligned} \Gamma_{n,\nu}(\alpha) &= \alpha \sum_{i=1}^{n} \rho_i - \sum_{i=1}^{n} D_\alpha(P_i\|Q_i) \\ &= \alpha \sum_{i=1}^{n} (\rho_i - \bar{\rho}_i) + \sum_{i=1}^{n} \Big\{ \alpha\bar{\rho}_i - D_\alpha(P_i\|Q_i) \Big\}. \end{aligned}$$

Since

$$\bar{\rho}_i = \sup_{\beta>1} \frac{D_\beta(P_i\|Q_i)}{\beta},$$

we have, for every fixed $\alpha > 1$,

$$D_\alpha(P_i\|Q_i) \leq \alpha\bar{\rho}_i.$$

Thus each term

$$\alpha\bar{\rho}_i - D_\alpha(P_i\|Q_i)$$

is nonnegative. Also $\rho_i - \bar{\rho}_i \geq 0$. Hence

$$\Gamma_{n,\nu}(\alpha) \geq 0.$$

If $\rho_i = \bar{\rho}_i$, the first sum vanishes, giving

$$\Gamma_{n,\nu}(\alpha) = \sum_{i=1}^{n} \Big\{ \alpha\rho_i - D_\alpha(P_i\|Q_i) \Big\}.$$

Finally, if

$$D_\alpha(P_i\|Q_i) = \alpha\rho_i \qquad \text{for all } i \text{ and all } \alpha > 1,$$

then each summand is zero, so

$$\Gamma_{n,\nu}(\alpha) = 0$$

for every $\alpha > 1$. This is the Gaussian-type equality case. Indeed, for an additive Gaussian mechanism with shift sensitivity Δ_i and variance σ_i^2, one has

$$D_\alpha(P_i\|Q_i) = \frac{\alpha\Delta_i^2}{2\sigma_i^2} = \alpha\rho_i,$$

where

$$\rho_i = \frac{\Delta_i^2}{2\sigma_i^2}.$$

□

5 Noise-Dependent Per-Coordinate Defect

For a fixed noise family ν, define the per-coordinate envelope defect at Renyi order $\alpha > 1$ by

$$g_\nu(\alpha; r) := \alpha r - D_\alpha(P_{\nu,r}\|Q_{\nu,r}),$$

where $(P_{\nu,r}, Q_{\nu,r})$ denotes a neighboring pair calibrated so that the exact zCDP radius is r. With this notation, under exact calibration,

$$\Gamma_{n,\nu}(\alpha) = \sum_{i=1}^{n} g_\nu(\alpha; \rho_i).$$

Thus the accounting defect is additive across coordinates, but the summand is not linear in ρ_i. The dependence on the allocation is therefore governed by the shape of $r \mapsto g_\nu(\alpha; r)$.

Assumption 3 (Local non-Gaussian envelope defect)**.** *Fix $\alpha > 1$. Assume that the noise family ν has a calibrated per-coordinate defect satisfying the following local expansion: there exist constants $r_0 > 0$, $c_\nu(\alpha) > 0$, and $C_\nu(\alpha) < \infty$ such that, for every $0 \leq r \leq r_0$,*

$$g_\nu(\alpha; r) \geq c_\nu(\alpha) r^2 - C_\nu(\alpha) r^3.$$

Equivalently, for sufficiently small r,

$$g_\nu(\alpha; r) \gtrsim_\nu r^2.$$

The constant $c_\nu(\alpha)$ is the leading non-Gaussian curvature defect of the Renyi curve relative to its zCDP envelope. For Gaussian noise,

$$g_\nu(\alpha; r) \equiv 0,$$

so $c_\nu(\alpha) = 0$.

Remark 2 (Interpretation through privacy-loss cumulants)**.** *Let*

$$L_t(x) = \log \frac{\mathrm{d}P_t}{\mathrm{d}Q_t}(x)$$

be the privacy-loss random variable, and let

$$K_t(\lambda) = \log \mathbb{E}_{Q_t} \exp\{\lambda L_t\}$$

be its cumulant generating function. Then

$$D_\alpha(P_t \| Q_t) = \frac{K_t(\alpha)}{\alpha - 1}.$$

The exact zCDP radius is

$$r(t) = \sup_{\beta > 1} \frac{K_t(\beta)}{\beta(\beta - 1)}.$$

Therefore

$$g_\nu(\alpha; r(t)) = \alpha r(t) - \frac{K_t(\alpha)}{\alpha - 1}.$$

Assumption 3 states that, after calibration by the exact zCDP radius, the cumulant generating function lies strictly below its best quadratic zCDP envelope by an amount of order at least r^2. Gaussian privacy loss is the exceptional case in which the cumulant generating function is exactly quadratic and the defect vanishes.

6 Allocation-Sensitive Lower Bounds

Corollary 1 (Allocation-sensitive lower bound)**.** *Suppose the assumptions of Theorem 1 hold and each coordinate is calibrated to its exact zCDP radius:*

$$\rho_i = \bar{\rho}_i.$$

Assume further that Assumption 3 holds at the fixed order $\alpha > 1$, and that

$$0 \le \rho_i \le r_* := \min\left\{ r_0, \frac{c_\nu(\alpha)}{2C_\nu(\alpha)} \right\}$$

for every i. Then

$$\Gamma_{n,\nu}(\alpha) \ge \frac{c_\nu(\alpha)}{2} \sum_{i=1}^n \rho_i^2.$$

Equivalently, writing

$$\rho = \sum_{i=1}^n \rho_i, \qquad w_i = \frac{\rho_i}{\rho}, \qquad H(w) = \sum_{i=1}^n w_i^2,$$

one has

$$\Gamma_{n,\nu}(\alpha) \geq \frac{c_\nu(\alpha)}{2}\rho^2 H(w).$$

Since

$$H(w) \geq \frac{1}{n},$$

the allocation minimizing the guaranteed defect for fixed total ρ is the uniform allocation

$$\rho_i = \frac{\rho}{n}.$$

In that case,

$$\Gamma_{n,\nu}(\alpha) \geq \frac{c_\nu(\alpha)}{2}\frac{\rho^2}{n}.$$

More generally, defining the effective number of compositions by

$$n_{\text{eff}} := \frac{\rho^2}{\sum_{i=1}^n \rho_i^2} = \frac{1}{H(w)},$$

one obtains

$$\Gamma_{n,\nu}(\alpha) \geq \frac{c_\nu(\alpha)}{2}\frac{\rho^2}{n_{\text{eff}}}.$$

Proof. By Theorem 1, under exact calibration,

$$\Gamma_{n,\nu}(\alpha) = \sum_{i=1}^n g_\nu(\alpha;\rho_i).$$

By Assumption 3,

$$g_\nu(\alpha;\rho_i) \geq c_\nu(\alpha)\rho_i^2 - C_\nu(\alpha)\rho_i^3.$$

Since

$$\rho_i \leq \frac{c_\nu(\alpha)}{2C_\nu(\alpha)},$$

we have

$$C_\nu(\alpha)\rho_i^3 \leq \frac{c_\nu(\alpha)}{2}\rho_i^2.$$

Therefore,

$$g_\nu(\alpha;\rho_i) \geq \frac{c_\nu(\alpha)}{2}\rho_i^2.$$

Summing over i yields

$$\Gamma_{n,\nu}(\alpha) \geq \frac{c_\nu(\alpha)}{2}\sum_{i=1}^n \rho_i^2.$$

Now write $\rho_i = \rho w_i$, where $\sum_i w_i = 1$. Then

$$\sum_{i=1}^n \rho_i^2 = \rho^2 \sum_{i=1}^n w_i^2 = \rho^2 H(w).$$

By Cauchy–Schwarz,

$$\left(\sum_{i=1}^n w_i\right)^2 \leq n\sum_{i=1}^n w_i^2.$$

Since $\sum_i w_i = 1$, this gives

$$H(w) = \sum_i w_i^2 \geq \frac{1}{n}.$$

Equality holds if and only if

$$w_1 = \cdots = w_n = \frac{1}{n},$$

equivalently

$$\rho_1 = \cdots = \rho_n = \frac{\rho}{n}.$$

Hence, for the uniform allocation,

$$\Gamma_{n,\nu}(\alpha) \geq \frac{c_\nu(\alpha)}{2}\frac{\rho^2}{n}.$$

Finally, the effective-composition form follows from

$$\sum_{i=1}^{n} \rho_i^2 = \frac{\rho^2}{n_{\text{eff}}}.$$

□

Proposition 1 (Convex envelope lower bound)**.** *Assume that, for the noise family ν and fixed order $\alpha > 1$, the function*

$$r \mapsto g_\nu(\alpha; r)$$

is convex on an interval containing $\rho_1, \ldots, \rho_n$. Then, for fixed total $\rho = \sum_i \rho_i$,

$$\Gamma_{n,\nu}(\alpha) = \sum_{i=1}^{n} g_\nu(\alpha; \rho_i) \geq n g_\nu\left(\alpha; \frac{\rho}{n}\right).$$

If, in addition,

$$g_\nu(\alpha; r) \geq a_\nu(\alpha) r^p$$

for some $p \geq 1$, then

$$\Gamma_{n,\nu}(\alpha) \geq a_\nu(\alpha) \sum_{i=1}^{n} \rho_i^p.$$

In particular, for $p = 2$,

$$\Gamma_{n,\nu}(\alpha) \geq a_\nu(\alpha)\frac{\rho^2}{n}.$$

Proof. The identity

$$\Gamma_{n,\nu}(\alpha) = \sum_{i=1}^{n} g_\nu(\alpha; \rho_i)$$

follows from Theorem 1 under exact calibration. If $g_\nu(\alpha; \cdot)$ is convex, Jensen's inequality gives

$$\frac{1}{n}\sum_{i=1}^{n} g_\nu(\alpha; \rho_i) \geq g_\nu\left(\alpha; \frac{1}{n}\sum_{i=1}^{n} \rho_i\right) = g_\nu\left(\alpha; \frac{\rho}{n}\right).$$

Multiplying by n gives

$$\Gamma_{n,\nu}(\alpha) \geq n g_\nu\left(\alpha; \frac{\rho}{n}\right).$$

If

$$g_\nu(\alpha; r) \geq a_\nu(\alpha) r^p,$$

then summing over i gives

$$\Gamma_{n,\nu}(\alpha) \geq a_\nu(\alpha) \sum_{i=1}^{n} \rho_i^p.$$

For $p = 2$, Cauchy–Schwarz gives

$$\sum_{i=1}^{n} \rho_i^2 \geq \frac{1}{n} \left(\sum_{i=1}^{n} \rho_i \right)^2 = \frac{\rho^2}{n}.$$

Thus

$$\Gamma_{n,\nu}(\alpha) \geq a_\nu(\alpha) \frac{\rho^2}{n}.$$

□

7 Loose Per-Coordinate Calibration

The previous bounds separate the intrinsic loss caused by replacing the full Renyi curve by a scalar zCDP radius from the extrinsic loss caused by certifying a coordinate with a loose zCDP parameter.

If $\rho_i \geq \bar{\rho}_i$, then Theorem 1 gives

$$\Gamma_{n,\nu}(\alpha) = \alpha \sum_{i=1}^{n} (\rho_i - \bar{\rho}_i) + \sum_{i=1}^{n} \Big\{ \alpha \bar{\rho}_i - D_\alpha(P_i \| Q_i) \Big\}.$$

Under the local defect assumption applied to $\bar{\rho}_i$, this yields

$$\Gamma_{n,\nu}(\alpha) \geq \alpha \sum_{i=1}^{n} (\rho_i - \bar{\rho}_i) + \frac{c_\nu(\alpha)}{2} \sum_{i=1}^{n} \bar{\rho}_i^2,$$

provided $\bar{\rho}_i \leq r_*$ for all i. The first term is calibration loss: it is present only because the certified parameter ρ_i is larger than the exact zCDP radius $\bar{\rho}_i$. The second term is intrinsic to the noise law and measures the departure of the exact Renyi curve from the Gaussian linear curve.

8 Conclusion

For an independently composed additive mechanism, the exact Renyi curve satisfies

$$D_\alpha(P \| Q) = \sum_{i=1}^{n} D_\alpha(P_i \| Q_i).$$

Therefore, under exact per-coordinate zCDP calibration,

$$\boxed{\Gamma_{n,\nu}(\alpha) = \sum_{i=1}^{n} \Big\{ \alpha \rho_i - D_\alpha(P_i \| Q_i) \Big\}.}$$

For Gaussian noise,

$$D_\alpha(P_i \| Q_i) = \alpha \rho_i,$$

so

$$\Gamma_{n,\nu}(\alpha) = 0$$

for every number of compositions n, every allocation $(\rho_1, \ldots, \rho_n)$, and every order $\alpha > 1$. Thus Gaussian zCDP accounting is exactly tight.

For a non-Gaussian smooth symmetric additive noise family satisfying the local curvature-defect assumption,

$$\boxed{\Gamma_{n,\nu}(\alpha) \gtrsim_\nu \sum_{i=1}^{n} \rho_i^2 = \rho^2 \sum_{i=1}^{n} \left(\frac{\rho_i}{\rho}\right)^2.}$$

Equivalently,

$$\boxed{\Gamma_{n,\nu}(\alpha) \gtrsim_\nu \frac{\rho^2}{n_{\text{eff}}}, \qquad n_{\text{eff}} = \frac{\rho^2}{\sum_i \rho_i^2}.}$$

The uniform allocation $\rho_i = \rho/n$ minimizes the guaranteed defect and gives

$$\Gamma_{n,\nu}(\alpha) \gtrsim_\nu \frac{\rho^2}{n}.$$

Highly nonuniform allocations decrease n_{eff}, increase $\sum_i \rho_i^2$, and therefore increase the suboptimality of scalar zCDP accounting relative to the exact Renyi privacy curve.